PROJECT REPORT

CSA2001: Fundamentals Of AI & ML
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Introduction

The rapid advancement of technology has significantly impacted the healthcare sector, enabling the use of Artificial Intelligence (AI) and Machine Learning (ML) to analyze and predict health-related conditions. Lifestyle factors such as diet, physical activity, sleep patterns, stress levels, and habits like smoking and alcohol consumption play a crucial role in determining an individual's health status.

However, many individuals are unaware of how these factors collectively influence their overall health and risk levels. This lack of awareness can lead to poor health decisions and increased chances of developing serious conditions.

The aim of this project is to develop a **Health Risk Predictor** using machine learning techniques that can estimate an individual's health risk level based on various input parameters. The system takes user inputs such as age, diet quality, exercise frequency, sleep duration, stress level, Body Mass Index (BMI), smoking habits, alcohol consumption, and family medical history to predict whether the individual falls under low, medium, or high health risk category.

This project demonstrates how machine learning can be applied to real-world problems and how predictive models can assist users in making informed decisions about their health and lifestyle.

Problem Statement

Many individuals do not monitor their health regularly and are unaware of how their daily lifestyle habits affect their overall health. Poor diet, lack of exercise, insufficient sleep, and high stress levels can increase health risks. However, there is no simple system available for users to analyze these factors together.

This project aims to develop a machine learning-based system that predicts health risk levels and helps users make informed lifestyle decisions.

Objectives

- To develop a machine learning model for predicting health risk
- To analyze the impact of lifestyle factors on health
- To create a CLI-based application for prediction
- To build a user-friendly interface using Streamlit
- To provide meaningful insights based on user input

Methodology

The project follows a machine learning-based approach. The dataset used in this project is obtained from Kaggle and contains various health-related attributes.

The data is preprocessed by converting categorical values into numerical format using encoding techniques. A machine learning model is then trained using this dataset.

The trained model is saved using a pickle file and later used in both CLI and UI applications to predict health risk based on user input.

Machine Learning Model

The model used in this project is a **classification algorithm** that predicts the health risk level as Low, Medium, or High based on input features. The model used is **Random Forest Classifier**, which is an ensemble learning algorithm.

A supervised learning approach is used, where the model is trained on labeled data containing health-related attributes and corresponding risk levels.

Random Forest is chosen because it provides high accuracy, handles both numerical and categorical data effectively, and reduces overfitting by combining multiple decision trees.

The algorithm learns patterns and relationships between input features such as diet, exercise, sleep, stress, BMI, and habits, and the output risk category.

The model works by creating multiple decision trees during training. Each tree makes a prediction, and the final output is determined based on the majority vote of all trees.

Once trained, the model is saved using a pickle file and used to make predictions on new user inputs.

Features such as BMI, stress level, and exercise frequency have a significant impact on the prediction results.

System Architecture

The system works by taking user inputs such as age, diet, exercise, sleep, stress level, BMI, smoking habits, alcohol consumption, and family history.

These inputs are processed and converted into numerical values, which are then passed to the trained machine learning model. The model analyzes the data and predicts the health risk level, which is displayed to the user.

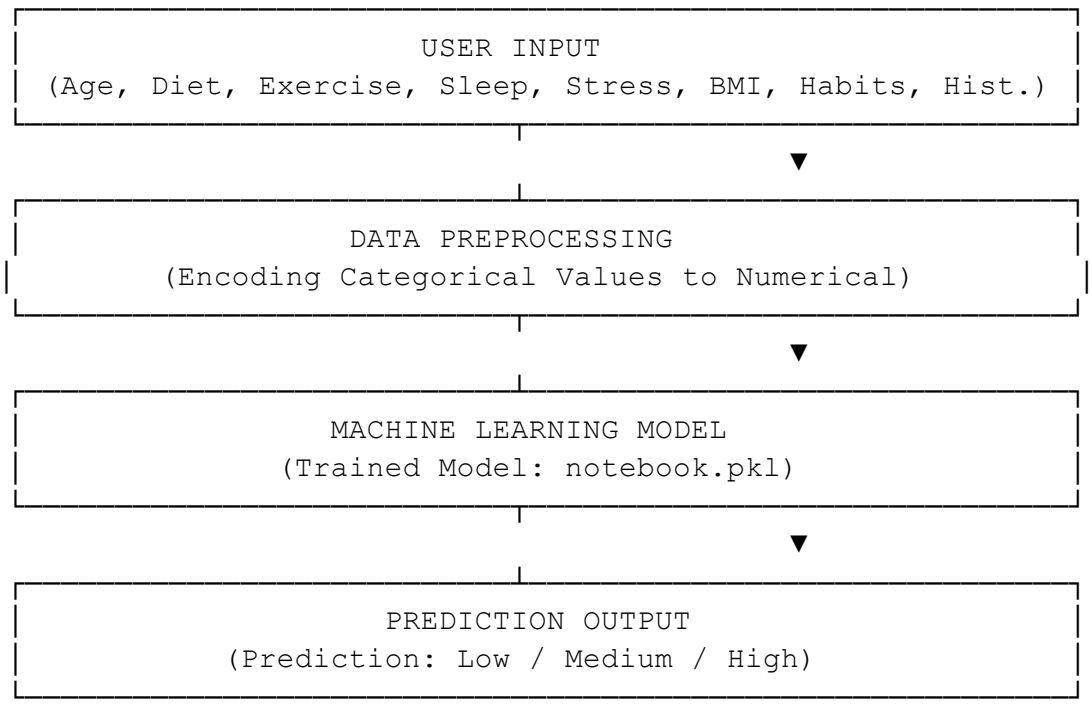

Figure 1: System Architecture of the Health Risk Predictor System

Implementation

The project is implemented using Python programming language.

The model is trained in a Jupyter Notebook and saved as a `.pkl` file. The CLI version is implemented using `main.py`, which takes input from the terminal and provides predictions.

Additionally, a Streamlit-based UI (`app.py`) is developed to provide an interactive interface for users.

Results

The system successfully predicts health risk levels based on user input. The predictions are displayed in both CLI and UI formats.

The results demonstrate that the model can effectively analyze lifestyle factors and classify risk levels. As shown in Figures 2 and 3, the system provides predictions through both CLI and user interface, ensuring flexibility and ease of use.

The model provides accurate and consistent predictions based on input parameters, demonstrating the effectiveness of machine learning in health risk analysis

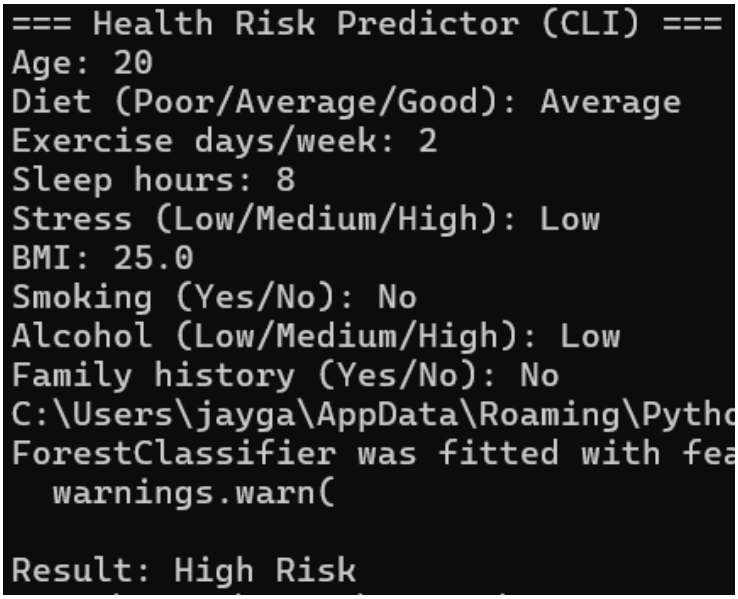

Figure 2: Command Line Interface Output of Health Risk Prediction

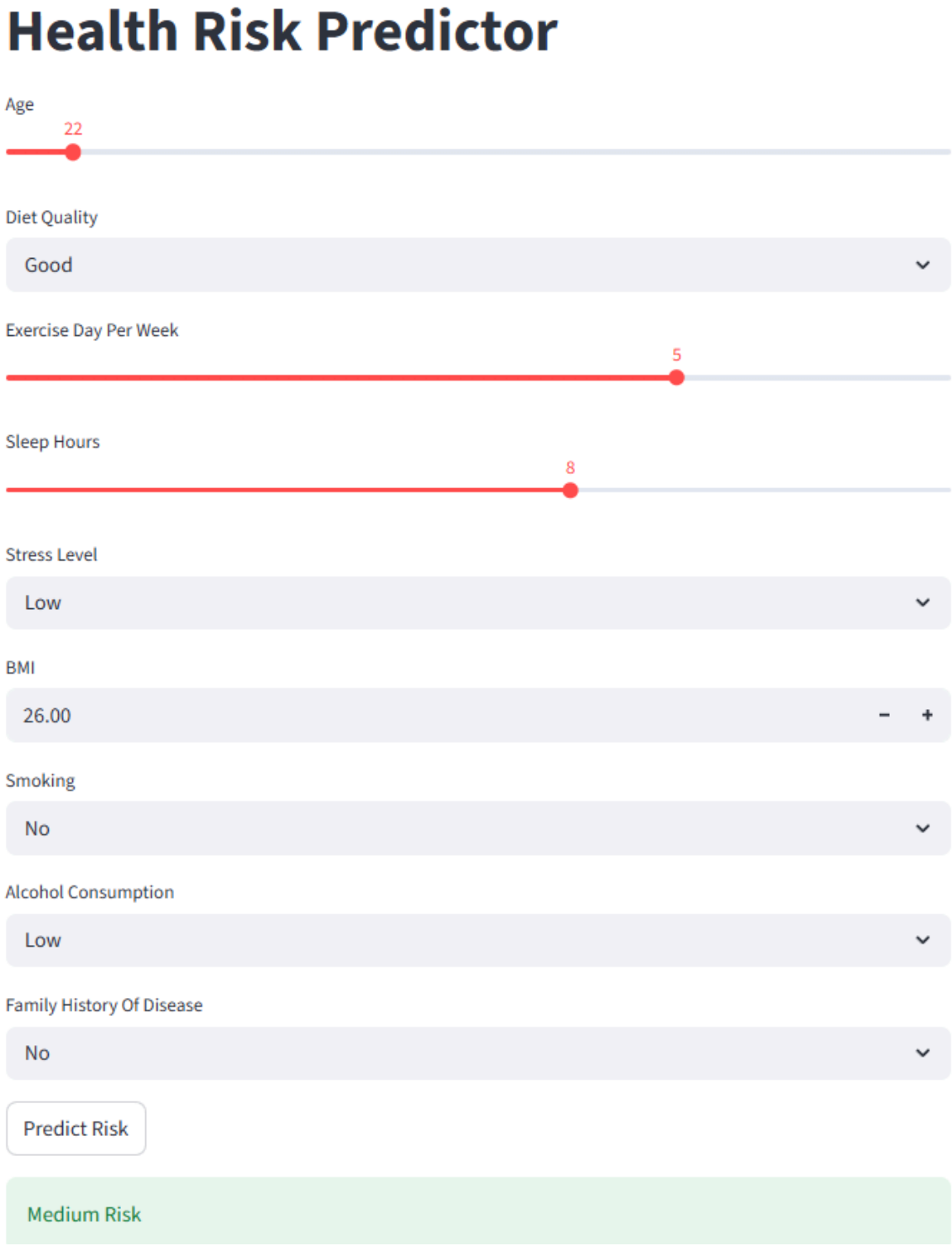

Figure 3: Streamlit-Based User Interface for Health Risk Prediction

```
model.fit(x_train,y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

```
from sklearn.metrics import accuracy_score
```

```
y_pred = model.predict(x_test)
```

```
acc = accuracy_score(y_pred,y_test)
```

```
print("Accuracy Score",acc)
```

```
Accuracy Score 0.96375
```

Figure 4: Accuracy Score of the Machine Learning Model

The model achieved an accuracy of approximately 96.375% on the dataset.

Advantages

- Easy to use
- Fast prediction
- Helps in health awareness
- Uses machine learning concepts
- Provides user-friendly interface

Limitations

- Uses dataset from Kaggle
- Not medically accurate
- Limited features
- Depends on input accuracy

Future Scope

- Use real healthcare datasets
- Improve model accuracy
- Add more health parameters
- Deploy as web/mobile application

Conclusion

This project demonstrates the use of machine learning in predicting health risks based on lifestyle factors. It shows how AI can be used to analyze user data and provide useful insights. The system can help individuals become more aware of their health and encourage better lifestyle choices.

This project also fulfills the requirement of CLI-based execution as specified in the course guidelines.